k230跌倒检测

k230和MSPM0通信

1. 实验前提

本教程使用的是MSPMG3507开发板，对应例程路径为【14.export\MSPM0-K230\10_k230_falldown_detect】。

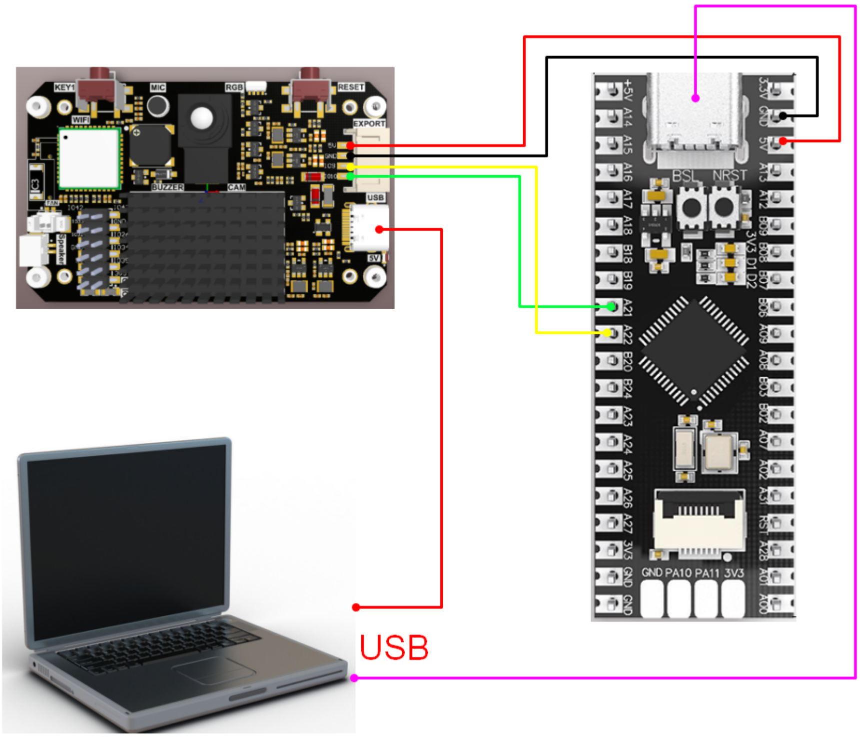
k230要运行【14.export\CanmvIDE-K230\10.falldown_detect.py】程序才能开始实验，建议下载为离线运行程序。

需要用到的物品：

Windows系统电脑
MSPMG3507开发板
K230视觉模块(包含烧录好镜像的TF卡)
两根type-C数据线
连接线

2. 实验接线

k230视觉模块	MSPM0开发板
5V	VCC
GND	GND
TXD(IO9)	A22
RXD(IO10)	A21



k230视觉模块	MSPM0开发板
5V	VCC
GND	GND
TXD(IO9)	A22
RXD(IO10)	A21

3. 主要代码讲解

```
void Pto_Data_Parse(uint8_t *data_buf, uint8_t num)
{
    uint8_t pto_head = data_buf[0];
    uint8_t pto_tail = data_buf[num-1];
    if (!(pto_head == PTO_HEAD && pto_tail == PTO_TAIL))
    {
        sprintf(print_buf, "pto error:pto_head=0x%02x , pto_tail=0x%02x\n",
pto_head, pto_tail);
        uart0_send_string(print_buf);
        return;
    }
    uint8_t data_index = 1;
    uint8_t field_index[PTO_BUF_LEN_MAX] = {0};
    int i = 0;
    int values[PTO_BUF_LEN_MAX] = {0};
    for (i = 1; i < num-1; i++)
    {
        if (data_buf[i] == ',')
        {
            data_buf[i] = 0;
            field_index[data_index] = i;
            data_index++;
        }
    }
    char msg[PTO_BUF_LEN_MAX] = {0};
    for (i = 0; i < data_index; i++)
    {
        if (i == 6)
        {
            memcpy(msg, (char*)data_buf+field_index[i]+1, field_index[i+1]-
field_index[i]);
        }
        else
```

```
        {
            values[i] = Pto_Char_To_Int((char*)data_buf+field_index[i]+1);
        }
    }

    uint8_t pto_len = values[0];

    if (pto_len != num)
    {
        sprintf(print_buf, "pto_len error:%d , data_len:%d\n", pto_len, num);
        uart0_send_string(print_buf);
        return;
    }
    uint8_t pto_id = values[1];
    if (pto_id != PTO_FUNC_ID)
    {
        sprintf(print_buf, "pto_id error:%d, func_id:%d\n", pto_id,
PTO_FUNC_ID);
        uart0_send_string(print_buf);
        return;
    }
    int x = values[2];
    int y = values[3];
    int w = values[4];
    int h = values[5];
    float score = values[7]/100.0;
    sprintf(print_buf, "falldown:x:%d, y:%d, w:%d, h:%d, state:'%s',
score:%.2f\n", x, y, w, h, msg, score);
    uart0_send_string(print_buf);
}
```

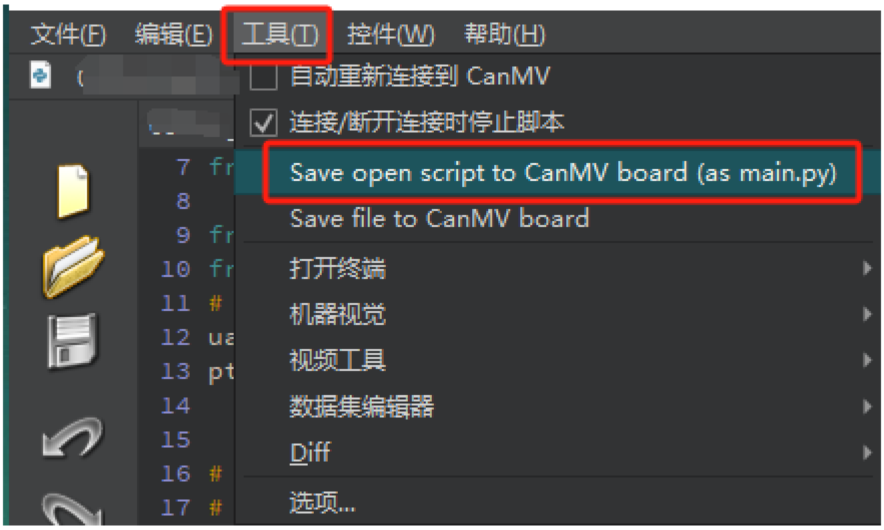
以上函数为解析K230数据的函数，只有符合特定协议才可以解析出对应的数据。

其中

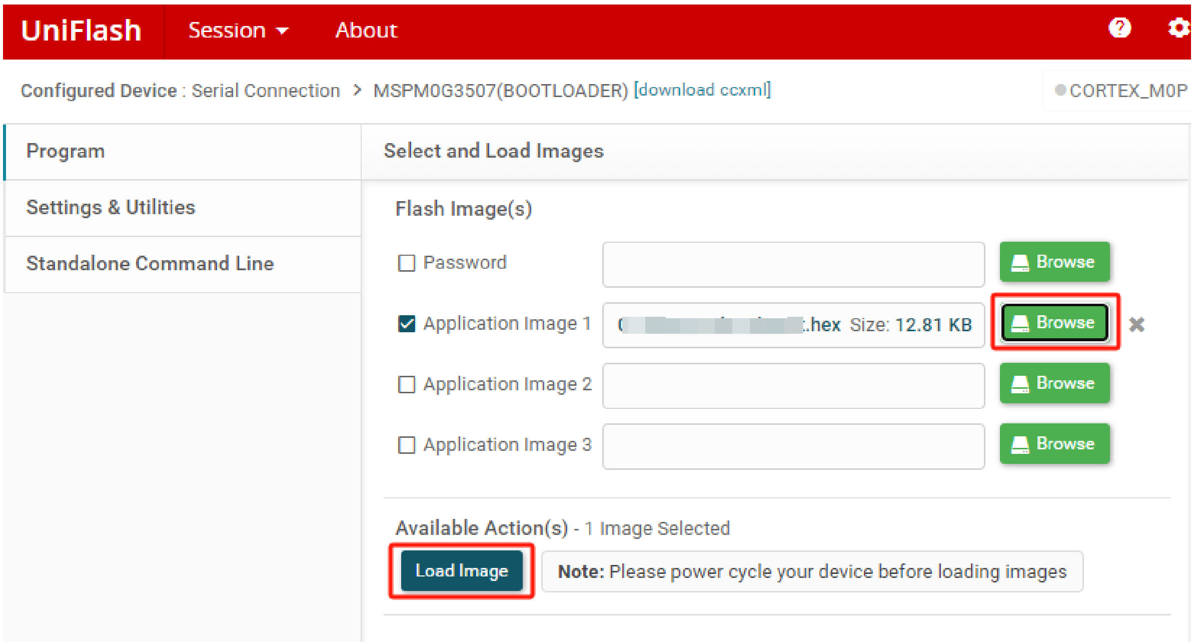
- x：是识别出来框左上角的横坐标
- y：是识别出来框左上角的纵坐标
- w：是识别出来框的宽度
- h：是识别出来框的长度
- state：是识别的状态，检测到跌倒状态是“Fall”，检测到未跌倒状态是“NoFall”
- score：是跌倒状态的分数

4. 实验现象

1. 连接好线后，k230视觉模块脱机运行
   K230连接Canmv IDE后，打开对应程序，点击工具栏的 【Save open script to CanMV board (as main.py)】，然后重启K230。



2. MSPM0烧录例程生成的hex文件。



3. 串口助手设置成如图的界面

4. 当K230摄像头画面识别到人体时，串口助手就会打印出k230传输给MSPM0的信息。

其中

- x：是识别出来框左上角的横坐标
- y：是识别出来框左上角的纵坐标
- w：是识别出来框的宽度
- h：是识别出来框的长度
- state：是识别的状态，检测到跌倒状态是“Fall”，检测到未跌倒状态是“NoFall”
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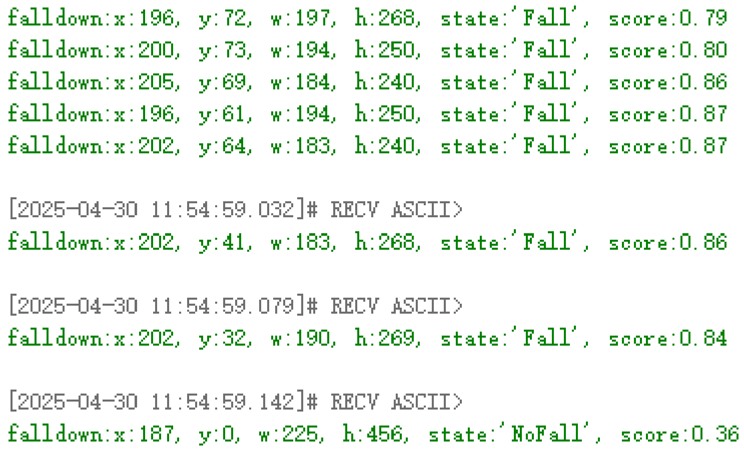
如下图所示



```
falldown:x:196, y:72, w:197, h:268, state:'Fall', score:0.79
falldown:x:200, y:73, w:194, h:250, state:'Fall', score:0.80
falldown:x:205, y:69, w:184, h:240, state:'Fall', score:0.86
falldown:x:196, y:61, w:194, h:250, state:'Fall', score:0.87
falldown:x:202, y:64, w:183, h:240, state:'Fall', score:0.87

[2025-04-30 11:54:59.032]# RECV ASCII>
falldown:x:202, y:41, w:183, h:268, state:'Fall', score:0.86

[2025-04-30 11:54:59.079]# RECV ASCII>
falldown:x:202, y:32, w:190, h:269, state:'Fall', score:0.84

[2025-04-30 11:54:59.142]# RECV ASCII>
falldown:x:187, y:0, w:225, h:456, state:'NoFall', score:0.36
```